

# AS1109 - Calculus & Differential Equations

## H2 Tutorial: Fresh Solved Example Set

25 September 2026

For students who joined recently

Every problem below is solved from the beginning. Before copying a calculation, first ask:

**What is the topic?** → **What rule applies?** → **What does the answer mean?**

Small sketches are included wherever they make the idea clearer.

### Q1. Domain and range from a geometric condition

#### Problem

Consider

$$f(x, y) = \sqrt{9 - x^2 - 4y^2}.$$

Find the **domain** and **range** of  $f$ . Also describe the domain geometrically.

#### Step-by-step solution

**Step 1: Find the domain.** A square root is real only when its inside is non-negative:

$$9 - x^2 - 4y^2 \geq 0.$$

Therefore

$$x^2 + 4y^2 \leq 9.$$

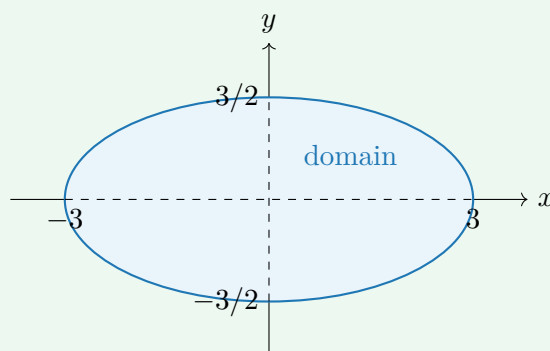
Write this in the standard ellipse form:

$$\frac{x^2}{9} + \frac{y^2}{9/4} \leq 1.$$

Hence

$$D = \{(x, y) : x^2 + 4y^2 \leq 9\}.$$

The ellipse has horizontal semi-axis 3 and vertical semi-axis  $3/2$ .



Because the boundary  $x^2 + 4y^2 = 9$  is included, the domain is **closed**. It fits inside a sufficiently large circle, so it is **bounded**.

**Step 2: Find the range.** Since a square root is non-negative,

$$f(x, y) \geq 0.$$

The smallest value is 0, attained on the ellipse boundary. The largest value occurs when  $x^2 + 4y^2$  is smallest, namely at  $(0, 0)$ :

$$f(0, 0) = \sqrt{9} = 3.$$

Therefore

$$\text{Range} = [0, 3].$$

**Quick memory cue**

$$\sqrt{\text{expression}} \Rightarrow \text{expression} \geq 0.$$

Domain questions often become familiar geometric regions after this first step.

**Q2. Open/closed and bounded/unbounded****Problem**

Classify the set

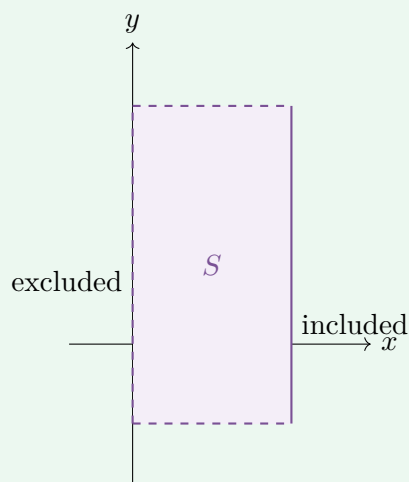
$$S = \{(x, y) : 0 < x \leq 2, -1 < y < 3\}$$

as **open**, **closed**, or **neither**, and as **bounded** or **unbounded**.

**Step-by-step solution**

The inequalities tell us exactly which boundary lines are included:

- $x = 0$  is **excluded** because  $0 < x$ ;
- $x = 2$  is **included** because  $x \leq 2$ ;
- $y = -1$  and  $y = 3$  are both **excluded**.



**Is  $S$  open?** No. Points on the boundary line  $x = 2$  belong to  $S$ , so not every point of  $S$  is an interior point.

**Is  $S$  closed?** No. For example, points of  $S$  can approach a point on  $x = 0$ , but that boundary point is not included.

Thus

$S$  is neither open nor closed.

**Is it bounded?** Yes. Both coordinates are restricted to finite intervals, so the whole set fits inside a sufficiently large disc.

$S$  is bounded.

**Important**

A set does **not** have to be either open or closed. Some sets are neither, and some sets can even be both in special spaces. For our present calculus examples, “neither” is a perfectly valid answer.

### Q3. Level curves: nested ellipses

#### Problem

For

$$f(x, y) = x^2 + 4y^2,$$

find and sketch the level curves corresponding to

$$c = 1, \quad c = 4, \quad c = 9.$$

#### Step-by-step solution

A level curve is obtained by setting

$$f(x, y) = c.$$

Therefore

$$x^2 + 4y^2 = c.$$

For  $c = 1$ :

$$x^2 + 4y^2 = 1 \quad \Longleftrightarrow \quad \frac{x^2}{1} + \frac{y^2}{1/4} = 1.$$

Semi-axes: 1 and  $1/2$ .

For  $c = 4$ :

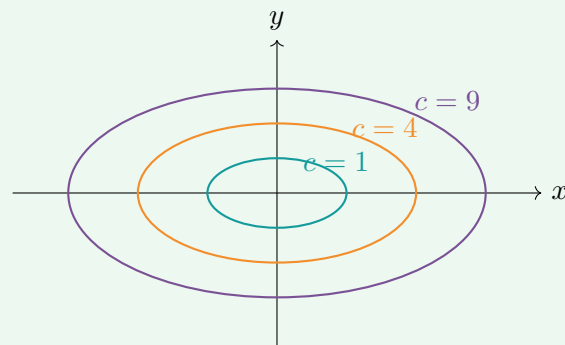
$$x^2 + 4y^2 = 4 \quad \Longleftrightarrow \quad \frac{x^2}{4} + \frac{y^2}{1} = 1.$$

Semi-axes: 2 and 1.

For  $c = 9$ :

$$x^2 + 4y^2 = 9 \quad \Longleftrightarrow \quad \frac{x^2}{9} + \frac{y^2}{9/4} = 1.$$

Semi-axes: 3 and  $3/2$ .



Hence the level curves are **nested ellipses** centred at the origin. As  $c$  increases, the ellipses expand.

#### Physical meaning

If  $f(x, y)$  were temperature, then each ellipse would be an **isotherm**: every point on one ellipse would have the same temperature.

## Q4. A multivariable limit that does not exist

### Problem

Determine whether

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$$

exists.

### Step-by-step solution

For a function of two variables, we may approach  $(0,0)$  along different paths. If two paths give different limiting values, the limit cannot exist.

**Path 1:**  $y = 0$ .

$$\frac{x(0)}{x^2 + 0} = 0.$$

Thus along  $y = 0$ ,

$$\lim f(x, y) = 0.$$

**Path 2:**  $y = x$ .

$$\frac{x \cdot x}{x^2 + x^2} = \frac{x^2}{2x^2} = \frac{1}{2} \quad (x \neq 0).$$

Thus along  $y = x$ ,

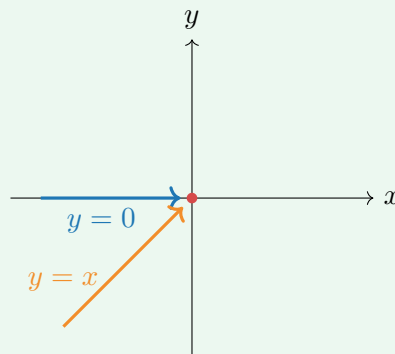
$$\lim f(x, y) = \frac{1}{2}.$$

Because

$$0 \neq \frac{1}{2},$$

the limit does not exist:

$$\boxed{\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2} \text{ does not exist.}}$$



### Exam logic

Two different path values are enough to prove that a limit **does not exist**. But checking two paths that happen to give the same value is **not enough** to prove that a limit exists.

## Q5. First and mixed partial derivatives

### Problem

Let

$$f(x, y) = x^2 e^y + y \sin x.$$

Find

$$f_x, \quad f_y, \quad f_{xy}.$$

Then verify that  $f_{xy} = f_{yx}$ .

### Step-by-step solution

**First, find  $f_x$ .** While differentiating with respect to  $x$ , treat  $y$  as a constant:

$$\frac{\partial}{\partial x}(x^2 e^y) = 2x e^y, \quad \frac{\partial}{\partial x}(y \sin x) = y \cos x.$$

Hence

$$f_x = 2x e^y + y \cos x.$$

**Now find  $f_y$ .** This time treat  $x$  as a constant:

$$\frac{\partial}{\partial y}(x^2 e^y) = x^2 e^y, \quad \frac{\partial}{\partial y}(y \sin x) = \sin x.$$

Therefore

$$f_y = x^2 e^y + \sin x.$$

**Find  $f_{xy}$ .** Differentiate  $f_x$  with respect to  $y$ :

$$f_{xy} = \frac{\partial}{\partial y}(2x e^y + y \cos x) = 2x e^y + \cos x.$$

So

$$f_{xy} = 2x e^y + \cos x.$$

**Check  $f_{yx}$ .** Differentiate  $f_y$  with respect to  $x$ :

$$f_{yx} = \frac{\partial}{\partial x}(x^2 e^y + \sin x) = 2x e^y + \cos x.$$

Therefore

$$f_{xy} = f_{yx}.$$

### Language cue

“Differentiate with respect to  $x$ ” means: **allow  $x$  to change and temporarily treat the other variables as constants.**

## Q6. Implicit differentiation without a circle

### Problem

For the curve

$$x^2 + xy + 2y^2 = 12,$$

find  $\frac{dy}{dx}$ .

### Step-by-step solution

The equation does not explicitly say  $y = \dots$ , but along the curve we may think of

$$y = y(x).$$

Therefore every time we differentiate a term involving  $y$ , the chain rule produces a factor  $dy/dx$ . Differentiate the entire equation with respect to  $x$ :

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(xy) + \frac{d}{dx}(2y^2) = \frac{d}{dx}(12).$$

Term by term:

$$2x + \left(x \frac{dy}{dx} + y\right) + 4y \frac{dy}{dx} = 0.$$

Collect the terms containing  $dy/dx$ :

$$(x + 4y) \frac{dy}{dx} + 2x + y = 0.$$

Hence

$$(x + 4y) \frac{dy}{dx} = -(2x + y),$$

so

$$\boxed{\frac{dy}{dx} = -\frac{2x + y}{x + 4y}}.$$

**Shortcut.** Define

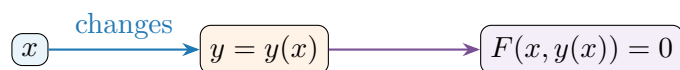
$$F(x, y) = x^2 + xy + 2y^2 - 12.$$

Then

$$F_x = 2x + y, \quad F_y = x + 4y,$$

and therefore

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{2x + y}{x + 4y}.$$



Implicit differentiation is simply the **chain rule applied to an equation that remains equal to 0**.

## Q7. Linearisation: approximating a difficult value

### Problem

Let

$$f(x, y) = \sqrt{4 + x - 2y}.$$

Find the linearisation of  $f$  at  $(0, 0)$  and use it to approximate

$$\sqrt{4.08}.$$

### Step-by-step solution

For a differentiable function  $f(x, y)$ , the linearisation at  $(a, b)$  is

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

Here  $(a, b) = (0, 0)$ .

First,

$$f(0, 0) = \sqrt{4} = 2.$$

Now

$$f_x = \frac{1}{2\sqrt{4 + x - 2y}}, \quad f_y = -\frac{1}{\sqrt{4 + x - 2y}}.$$

At  $(0, 0)$ ,

$$f_x(0, 0) = \frac{1}{4}, \quad f_y(0, 0) = -\frac{1}{2}.$$

Therefore

$$L(x, y) = 2 + \frac{x}{4} - \frac{y}{2}.$$

To approximate  $\sqrt{4.08}$ , choose  $x = 0.08$  and  $y = 0$ :

$$\sqrt{4.08} = f(0.08, 0) \approx L(0.08, 0).$$

Thus

$$L(0.08, 0) = 2 + \frac{0.08}{4} = 2 + 0.02 = 2.02.$$

Hence

$$\sqrt{4.08} \approx 2.02.$$

### Geometric idea

Near a point, a differentiable surface is well approximated by its **tangent plane**. Linearisation replaces a complicated function by that simpler plane for nearby inputs.

## Q8. Engineering application of the multivariable chain rule

### Problem

The temperature in a metal plate is

$$T(x, y) = 50 + x^2 + 2y^2.$$

A small sensor moves according to

$$x = t, \quad y = t^2.$$

Find the rate at which the temperature experienced by the sensor changes at  $t = 1$ .

### Step-by-step solution

The temperature depends on position  $(x, y)$ , but the sensor's position depends on time  $t$ :

$$T = T(x, y), \quad x = x(t), \quad y = y(t).$$

Therefore use the multivariable chain rule:

$$\frac{dT}{dt} = T_x \frac{dx}{dt} + T_y \frac{dy}{dt}.$$

Compute the required derivatives:

$$\begin{aligned} T_x &= 2x, & T_y &= 4y, \\ \frac{dx}{dt} &= 1, & \frac{dy}{dt} &= 2t. \end{aligned}$$

Hence

$$\frac{dT}{dt} = 2x(1) + 4y(2t).$$

Since  $x = t$  and  $y = t^2$ ,

$$\frac{dT}{dt} = 2t + 8t^3.$$

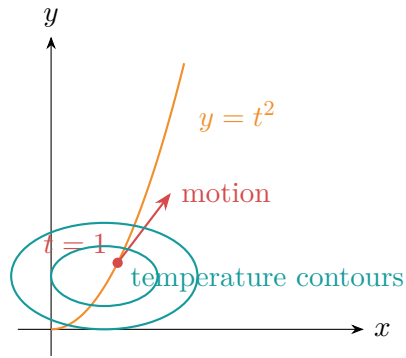
At  $t = 1$ ,

$$\left. \frac{dT}{dt} \right|_{t=1} = 2 + 8 = 10.$$

So the sensor is experiencing an **increase** in temperature at a rate of

$$10 \text{ temperature-units per unit time.}$$

The positive sign is physically meaningful: at that instant the sensor is moving into a hotter region.



## One-page recap

### Core ideas to remember

| Topic                    | Key idea                                                                                                       |
|--------------------------|----------------------------------------------------------------------------------------------------------------|
| Domain                   | Make the formula meaningful: square roots need $\geq 0$ , logarithms need $> 0$ , denominators need $\neq 0$ . |
| Open/closed              | Open: every point is interior. Closed: contains all boundary points.                                           |
| Bounded                  | The set fits inside some sufficiently large disc.                                                              |
| Level curve              | Set $f(x, y) = c$ and identify/sketch the resulting curve.                                                     |
| Two-variable limit       | Different path values $\Rightarrow$ the limit does not exist.                                                  |
| Partial derivative       | Differentiate with respect to one variable while treating the others as constants.                             |
| Implicit differentiation | Remember that $y = y(x)$ and apply the chain rule to the whole equation.                                       |
| Linearisation            | Near a point, replace the surface by its tangent plane.                                                        |
| Chain rule               | Multiply derivatives along each dependency path and add the path contributions.                                |

End of solved example set